TISSUES

📝 Part 1: In-Chapter Point-Wise Notes

🤝 3.1 Tissues — "The Teams of Workers"

- A **tissue** is a group of similar cells performing a specific function.
- **The Biological Hierarchy:**
    - Several tissues working together form an **organ** ➡️ *(Example: Tongue having epithelial, nerve, and muscle cells).*
    - Many organs acting together perform a specific life process forming an **organ system** ➡️ *(Example: Digestive system).*
    - Organ systems together constitute the **organism** ➡️ *(Example: The root system and shoot system constitute a plant).*

🌿 3.2 Plant Tissues

- Plant tissues are fundamentally classified into two main types:

Parameter 📏	**Meristematic Tissue** 🟢	**Permanent Tissue** 🟤
Cell Division	Actively dividing (cells can multiply to produce new cells).	Non-dividing (lost their ability to divide).
Specialization	Unspecialised cells.	Specialised cells taking a permanent shape.
Living Status	Always living.	May be living (nucleus present) or dead (nucleus absent).

🌱 A. Meristematic Tissue or Meristem

- **Chief Characteristics of Meristematic Tissue:**
    1. Cells are small and usually cubical.

2. Cell wall is thin.
3. Nuclei are large.
4. Vacuoles are almost absent (because there is no need for storage).
5. Cells are tightly packed with almost no intercellular spaces.
6. New cells produced are eventually transformed into mature permanent tissues.

- **Types of Meristematic Tissue:**
  - **(i) Apical or Terminal Meristem:**
    - 📍 **Location:** Tips of roots and stems, growing young leaves, and tips of axillary buds.
    - ⚙️ **Function:** Adds new cells to increase the length of the plant.
  - **(ii) Lateral or Cambium Meristem:**
    - 📍 **Location:** Situated below the bark.
    - ⚙️ **Function:** Responsible for the increase in girth (diameter) of the stem.

🧱 B. Permanent Tissues

- According to the function performed, permanent tissues are classified into three types: Protective, Supporting, and Conducting.

🛡️ 1. Protective Tissue

- Consists of cells with thick walls.
- 📍 **Location:** Found on the surface of roots, stems, and leaves.
- **Inline Examples:**
  - ➡️ *(Example: Epidermis of leaves secretes a waxy water-proof material called cuticle).*
  - ➡️ *(Example: Cork cells in tree barks contain another strong water-proof material).*

📐 2. Supporting Tissue (Simple Tissues)

- Simple tissues are made up of only one type of cell. There are three most important types:

Parameter 📏	(a) Parenchyma 🟢	(b) Collenchyma (collen: glue) 🟡	(c) Sclerenchyma (scleros: hard) 🟤
Cell Structure	Large, thin-walled, oval/circular/polygonal cells with a single large vacuole.	Elongated cells with cell walls thickened at the corners.	Long, narrow, **dead** cells with very thick walls due to lignin deposition.
📍 **Location**	Soft parts of plants (cortex/outer region and pith/central region).	Leaf stalks and below the epidermis of stems.	Stems and veins of the leaves.
⚙️ **Function**	Temporary support; May store food ➡️ *(Example: Potatoes).*	Helps to support the parts of a plant (provides flexibility).	Provides hard strength to plant parts.

- **Specialized Parenchyma (Chlorenchyma):** Parenchyma cells in the leaf containing chloroplasts are termed chlorenchyma. ⚙️ **Function:** Helps the leaf to produce food by photosynthesis.
- **Sclerenchyma Fibres and Stones:**
    - ➡️ *(Example: Linen and hemp ropes/mats are made of sclerenchyma plant fibres).*

- ➡️ *(Example: Walnut shells and gritty masses in pear skin/pulp are made of sclerenchyma "stone cells").*

💧 3. Conducting Tissue (Complex / Vascular Tissue)

- Complex tissues are made of more than one type of cells working together as a unit.
- ⚙️ **Function:** Provides a passage for water and dissolved materials to move up and down in the plant.
- Vascular bundles are formed collectively by Xylem and Phloem ➡️ *(Example: Veins of leaves are vascular bundles).*

⬆️ (a) Xylem (Water Conducting)

- Cells are elongated and thick-walled, somewhat in the form of tubular passages.
- ⚙️ **Function:** Provides for the **upward movement** of water and dissolved materials absorbed from roots to other plant parts.
- *Note:* Older xylem (*xylo*: wood) forms the wood and does not participate in transport.
- **Components of Xylem:**
    1. **Tracheids:** Elongated dead cells with highly lignified walls and large cavities, placed end to end. ⚙️ **Function:** Provide mechanical support.
    2. **Xylem Vessels (Tracheae):** Long tube-like structures of elongated cells. Transverse walls get dissolved to make a continuous water pipe. ⚙️ **Function:** Transporting water and dissolved minerals.
    3. **Xylem Parenchyma:** Living cells. ⚙️ **Function:** Serve for the storage of food (sugars/starch) and help conduct water/minerals.
    4. **Xylem Fibres:** Dead sclerenchyma cells, long/narrow with tapering ends. ⚙️ **Function:** Provide mechanical support to the plant.

⬇️ (b) Phloem (Food Conducting)

- (*phloos*: bark)

- ⚙️ **Function:** Provides a passage for the **downward movement** of manufactured food from leaves to various parts, and **upward movement** to growing new leaves.
- **Components of Phloem:**
    1. **Sieve Tubes:** Elongated cells placed end to end forming a long tube. Transverse walls (sieve plates) are perforated. ⚙️ **Function:** Transport food from leaves to storage organs.
    2. **Companion Cells:** Living parenchyma cells closely associated with sieve tubes. ⚙️ **Function:** Help in the functioning of the sieve tube cells.
    3. **Phloem Parenchyma:** Parenchymatous tissue. ⚙️ **Function:** Storage of starch, fat, and other organic food material.
    4. **Phloem Fibres:** Sclerenchymatous cells. ⚙️ **Function:** Mainly provide support.

📝 Part 1: In-Chapter Point-Wise Notes

🐾 3.3 Animal Tissues

- The main kinds of animal tissues are: Epithelial, Connective, Muscle, and Nervous tissue.

🛡️ 1. Epithelial Tissue

- It is a thin, protective continuous sheet of cells.
- The cells fit tightly together leaving absolutely no space between them.
- Cell shapes may be flat, cuboidal, or columnar.
- ⚙️ **General Functions:** Functions for protection (as on skin), absorption and secretion (as in intestine), and sensory perception.
- 📍 **General Location:** Covers the surface of the body and lines various body cavities and internal organs, including blood vessels. ➡️ *(Example: Outermost layer of skin, lining surfaces of mouth, nose, lungs, stomach).*

Epithelial Tissue Type 🔬	Key Characteristics 📝	Exact Anatomical Location 📍	Primary Function ⚙️
(a) Squamous	Composed of thin, flat cells with prominent nuclei, closely packed, and irregular in shape.	Lining the mouth, nasal cavities, blood vessels, and lymph vessels.	Protects underlying parts from mechanical injury, germs, harmful chemicals, and drying up.
(b) Cuboidal	Cells are cuboidal in shape.	Some parts of kidney tubules and in some glandular ducts ➡️ *(Example: Salivary glands, pancreatic duct).*	Secretion and absorption (implied by location).
(c) Columnar	Vertically arranged cylindrical or brick-like cells, usually tall in size.	Inner lining of the stomach and intestines.	Absorption and secretion.
(i) Ciliated Columnar	Columnar cells possessing thread-like protoplasmic projections called cilia at their free ends.	Trachea (wind pipe).	Cilia constantly keep lashing to move materials entering these regions.

Epithelial Tissue Type 🔬	Key Characteristics 📝	Exact Anatomical Location 📍	Primary Function ⚙️
(ii) Glandular	A kind of columnar epithelium containing large cells. Folded inward to form compact, hollow, or tubular glands.	Lining of the stomach and intestine; ➡️ *(Example: Sweat glands, tear glands, liver).*	Secrete certain chemical substances.
(d) Stratified	Composed of several layers (strata) of the same or different kinds of epithelial cells. Upper layers are squamous, lowermost layer is cuboidal.	Found in the skin and the cornea.	Multi-layered protection.

🔗 **2. Connective Tissue**

- **Three Key Characteristics:** Contains an abundance of intercellular substance (the matrix), fewer cellular elements, and fibres.
- ⚙️ **General Function:** Binds one tissue with another and connects various organs, keeping them in their proper places.
- **Classification of Connective Tissue:**
    - **I. Connective Tissue Proper (Serves for packing and binding):**
        - **(a) Areolar (packing) tissue:** Most widely spread.
            - 📍 **Location:** Beneath the epidermis of skin.
            - ⚙️ **Function:** Makes the skin elastic and helps it to withstand pulling strain.

- **(b) Adipose (fat) tissue:** Has specialized cells which store fat.
    - 📍 **Location:** Padding under the skin and around kidneys, eye ball, etc.
    - ⚙️ **Function:** Padding acts as insulation for retaining body heat.
- **(c) Fibrous connective tissue:** Made up of elongated cells called fibres, bundled together by areolar tissue.
    - ⚙️ **Function:** Forms **tendons** (connect muscle to bone) and **ligaments** (connect bone to bone and hold them in position).

- **II. Supportive Connective Tissue (Gives support):**
    - **(a) Cartilage:** A non-porous tissue with thickened matrix. It is semi-transparent and elastic, containing NO blood vessels or nerves.
        - 📍 **Location:** Tip of nose, external ears, trachea, bronchial tubes, between vertebrae, and ends of long bones.
    - **(b) Bone:** A hard porous tissue with a good supply of blood vessels and nerves. Consists of living cells (called osteoblasts or osteocytes) and a rigid mass of inorganic salts like Ca and P.

- **III. Fluid Connective Tissue (Blood and Lymph):**
    - **(a) Blood:** Composed of a liquid part (plasma) and a cellular part (red blood cells, white blood cells, and platelets).
    - **(b) Lymph:** Fluid surrounding body cells, essentially blood plasma that has oozed out of blood vessels. Contains white blood cells but NOT red blood cells.
    - ⚙️ **Function (Blood & Lymph):** Mainly concerned with transportation (including respiratory gases) and immunity (protection against diseases).

💪 3. Muscle Tissue

- Forms the muscles of the body.
- ⚙️ **Function:** Muscles can contract and relax, thus helping the body in its movements.

Parameter 📏	**(a) Striated (Skeletal/Striped)** 🔴	**(b) Unstriated (Smooth/Unstriped)** ⚪	**(c) Cardiac** ❤️
Control	Voluntary (under control of our will). Moves only when you want it to.	Involuntary (not under the control of our will).	Involuntary in function. Contracts without outside stimulation.
Cell Structure	Made of long fibres (cells), nucleated, and striated (light and dark bands run across fibres).	Spindle-shaped/slender tapering cells, uninucleated.	Striated, uninucleated, and branched. Relatively short.
📍 **Location**	Arms, legs, face, neck, etc. (Constitutes about 50% of body weight).	Walls of intestine, muscles of the iris of the eye, skin, lining of blood vessels, urinary bladder, uterus, etc..	Found only in the walls of the heart.
Fatigue Level	Gets tired.	Standard fatigue.	Do not get tired soon.

🧠 4. Nervous or Neural Tissue

- Constitutes the nervous system and is made up of specialised elongated cells called **neurons** (nerve cells).
- **Structure of a Neuron:**
    - **Cell Body:** Called perikaryon or cyton, which contains the nucleus.
    - **Dendrites/Dendrons:** One or more elongated hair-like extensions attached to the cell body.
    - **Axon:** One specialised extension that may be very long (sometimes one metre, or even three metres as in an elephant).
- **Nerves:** Axons bundled together form a nerve.
- ⚙️ **Role/Function:** Concerned with the perception and responses of animals.

🏆 Part 2: End-of-Chapter Master Lists (Tissues)

📚 1. Combined Terms List (Glossary)

Biological Term 🔑	Exam-Ready Definition 🧠
Tissue	A group of similar cells performing a **specific function**.
Organ	Several tissues acting together contributing to some **specific function** inside the body.
Organ System	Many organs acting together to perform a **specific life process**.
Meristematic Tissue	Plant tissue made of actively dividing cells that multiply to **produce new cells**.
Permanent Tissue	Plant tissue made of specialised, non-dividing cells that take a **permanent shape**.

Biological Term 🔑	Exam-Ready Definition 🧠
Chlorenchyma	Parenchymatous cells in the leaf that contain chloroplasts to produce food by **photosynthesis**.
Tracheids	Elongated dead xylem cells with highly lignified walls and large cavities that provide **mechanical support**.
Xylem Vessels (Tracheae)	Long tube-like structures of elongated cells with dissolved transverse walls meant for **transporting water**.
Sieve Tubes	Elongated phloem cells placed end to end with perforated transverse walls to **transport food**.
Companion Cells	Living parenchyma cells closely associated with sieve tubes that help in the functioning of **sieve tube cells**.
Vascular Bundles	Complex conducting tissues collectively formed by the **xylem and phloem**.
Cilia	Thread-like protoplasmic projections at the free ends of certain columnar epithelial cells that **keep lashing** to move materials.
Matrix	The abundant intercellular substance found in **connective tissues**.
Tendons	Fibrous connective tissue that connects **muscle to bone**.

Biological Term 🔑	Exam-Ready Definition 🧠
Ligaments	Fibrous connective tissue that connects **bone to bone** and holds them in position.
Lymph	Fluid surrounding body cells, essentially blood plasma without red blood cells, concerned with **transportation and immunity**.
Neuron	A specialised elongated cell that makes up the **nervous tissue**.
Perikaryon (Cyton)	The cell body of a neuron containing the **nucleus**.
Nerve	A bundle of **axons** grouped together.

🌟 2. Master Example List

- **Biological Hierarchy:**
  - *Organ:* Tongue (has epithelial, nerve, muscle cells).
  - *Organ System:* Digestive system, root system, shoot system.
- **Plant Tissue Modifications:**
  - *Protective Tissue:* Epidermis of leaves (cuticle), cork cells in tree barks.
  - *Parenchyma storing food:* Potatoes.
  - *Sclerenchyma Fibres:* Linen and hemp ropes/mats.
  - *Sclerenchyma "Stone cells":* Walnut shells, gritty masses in pear skin and pulp.
  - *Vascular Bundles:* Veins of leaves.

- **Animal Epithelial Tissue Locations:**
    - *Cuboidal Epithelium (Glandular ducts):* Salivary glands, pancreatic duct.
    - *Glandular Epithelium (Inward folds):* Sweat glands, tear glands, liver.

⚙️ 3. Master Function & Location Matrix

Biological Part 🧬	Exact Anatomical Location 📍	Primary Function ⚙️
Apical (Terminal) Meristem	Tips of roots/stems, growing young leaves, tips of axillary buds.	Adds new cells to increase length.
Lateral (Cambium) Meristem	Situated below the bark.	Responsible for the increase in girth (diameter) of the stem.
Protective Tissue	Surface of roots, stems, and leaves.	Protects plant; secretes water-proof materials.
Parenchyma	Soft parts of plants (cortex/outer region, pith/central region).	Temporary support; stores food.
Collenchyma	Leaf stalks and below the epidermis of stems.	Helps support plant parts (provides flexibility).
Sclerenchyma	Stems and veins of the leaves.	Provides hard strength to plant parts.

Biological Part 🧬	Exact Anatomical Location 📍	Primary Function ⚙️
Xylem	Vascular bundles (roots, stems, leaves).	Upward movement of water and dissolved materials from roots.
Phloem	Vascular bundles (roots, stems, leaves).	Downward/upward movement of manufactured food from leaves.
Squamous Epithelium	Lining the mouth, nasal cavities, blood vessels, lymph vessels.	Protects underlying parts from injury, germs, drying up.
Cuboidal Epithelium	Kidney tubules, some glandular ducts.	Secretion and absorption.
Columnar Epithelium	Inner lining of the stomach and intestines.	Absorption and secretion.
Ciliated Columnar Epithelium	Trachea (wind pipe).	Cilia lash to move entering materials.
Stratified Epithelium	Skin and the cornea.	Multi-layered protection.
Areolar (Packing) Tissue	Beneath the epidermis of skin.	Makes skin elastic; helps withstand pulling strain.

Biological Part 🧬	Exact Anatomical Location 📍	Primary Function ⚙️
Adipose (Fat) Tissue	Padding under skin, around kidneys, eye ball, etc.	Stores fat; padding acts as insulation for retaining body heat.
Cartilage	Tip of nose, external ears, trachea, bronchial tubes, ends of long bones.	Provides flexible, semi-transparent support.
Bone	The skeleton (arms, legs, etc.).	Hard, rigid mass providing structure and support.
Blood & Lymph	Blood vessels and spaces surrounding body cells.	Transportation of materials and immunity against diseases.
Striated (Voluntary) Muscle	Arms, legs, face, neck, etc.	Voluntary movement (moves when you want it to).
Unstriated (Involuntary) Muscle	Walls of intestine, iris of eye, skin, lining of blood vessels, bladder.	Involuntary movement (not under control of our will).
Cardiac Muscle	Walls of the heart.	Involuntary rhythmic contraction (does not get tired).
Nervous (Neural) Tissue	Brain, spinal cord, nerves.	Perception and responses of animals.

📘 C. SHORT ANSWER TYPE

1. Name any one body part where ciliated epithelium is found in humans? What is its function?

- **Body Part:** Found in the inner lining of the windpipe (trachea).
- ⚙️ **Function:** The cilia constantly keep lashing to move materials (like mucus and trapped dust) that enter these regions.

2. Define the terms:

- **(a) Tissue:** A group of similar cells performing a specific function.
- **(b) Organ:** Several tissues acting together contributing to some specific function inside the body.
- **(c) Organ system:** Many organs acting together to perform a specific life process.
- **(d) Lymph:** The fluid surrounding body cells; it is essentially blood plasma that has oozed out of the blood vessels.
- **(e) Neuron:** A specialised elongated cell that makes up the nervous tissue.

3. Write the specific locations of the following tissues in plants/animals.

- **(a) Sclerenchyma:** Stems and veins of the leaves.
- **(b) Collenchyma:** Leaf stalks and below the epidermis of stems.
- **(c) Ciliated columnar epithelium:** Trachea (windpipe).
- **(d) Cardiac muscles:** Walls of the heart.

4. List the tissues found in the human heart?

- Cardiac muscle tissue, Epithelial tissue, Nervous tissue, and Connective tissue (blood).

5. Can you consider a cluster of eggs as a tissue? Why?

- **No**, because they lack the defining characteristics of a tissue. A tissue is made of structurally similar cells working together to perform a specific function, which a random cluster of eggs does not do.

6. Name the three kinds of muscles found in the human body. In each case, name one region in the body where they are found.

- **Striated (Voluntary) muscles:** Arms or legs.
- **Unstriated (Involuntary) muscles:** Walls of the intestine or urinary bladder.
- **Cardiac muscles:** Walls of the heart.

📕 D. LONG ANSWER TYPE

1. Distinguish between the following pairs with reference to what is asked in the brackets.

Pair ⚖️	Difference based on specific parameter 📏
(a) Parenchyma vs. Collenchyma *(cellular structure)*	**Parenchyma** has large, thin-walled cells. **Collenchyma** cells are elongated with walls thickened at the corners.
(b) Meristematic vs. Permanent tissue *(cell-division)*	**Meristematic** cells actively divide to produce new cells. **Permanent** cells are non-dividing (have lost their ability to divide).
(c) Blood vs. Lymph *(cellular components)*	**Blood** contains red blood cells, white blood cells, and platelets. **Lymph** contains white blood cells but *no* red blood cells.
(d) Bone vs. Cartilage *(porosity)*	**Bone** is a hard, *porous* tissue. **Cartilage** is a *non-porous* tissue.

Pair ⚖️	Difference based on specific parameter 📏
(e) Voluntary vs. Involuntary muscles *(shape and cellular structure)*	**Voluntary** are long, multi-nucleated, and striated. **Involuntary** are spindle-shaped/tapering, uninucleated, and unstriated.

2. What is the difference between:

Pair ⚖️	Key Difference 📏
(a) Cell and Tissue	A **cell** is the fundamental unit of life. A **tissue** is a group of similar cells working together.
(b) Organ and Organism	An **organ** is a part of the body made of tissues. An **organism** is the complete living being made of organ systems.
(c) Organ and Organelle	An **organ** is a macroscopic body part. An **organelle** is a microscopic living part *inside* a cell with a definite function.
(d) Organ and Organ system	An **organ** performs a specific function. An **organ system** is a group of many organs acting together for a major life process.

🟢 E. STRUCTURED/APPLICATION QUESTIONS

1. Study the diagram given alongside (Phloem) and then answer:

- **(a) Identify the tissue and give a reason to support your answer.**
    - **Phloem tissue.** Reason: It is a complex tissue showing sieve tubes and companion cells working together.
- **(b) Name the parts labelled 1, 2, 3, and 4.**

    - 1: Sieve plate, 2: Sieve tube, 3: Phloem parenchyma, 4: Companion cell.
- **(c) Where is this tissue likely to be found in the plant?**
    - In the vascular bundles of roots, stems, and leaves.
- **(d) State the function of the parts labelled 1, 2, 3 and 4.**
    - 1 (Sieve plate): Allows passage of food. 2 (Sieve tube): Transports food. 3 (Phloem parenchyma): Stores organic food. 4 (Companion cell): Helps the sieve tube function.

2. Study the diagram given below (Neuron) and then answer:

- **(a) Identify the cell.**
    - Neuron (Nerve cell).
- **(b) Name the parts labelled 1, 2, 3, 4, 5, and 6.**
    - 1: Axon endings, 2: Axon, 3: Dendrite, 4: Cyton (Perikaryon), 5: Nucleus, 6: Node of Ranvier.
- **(c) Where is this cell likely to be found in the human body and what is its function?**
    - Found in the brain/nervous system of an animal. ⚙️ **Function:** Perception and response to stimuli.

3. The figures (A and B) given ahead are two kinds of animal tissues:

- **(a) Identify the tissues A and B.**
    - A = Cartilage. B = Bone.
- **(b) Label all the guidelines marked in figures A and B.**
    - A: 1 = Matrix, 2 = Empty Lacuna / Cartilage cell.
    - B: 1 = Haversian canal, 2 = Concentric lamellae, 3 = Bone cell.
- **(c) Write one common function of both the tissues A and B.**
    - Both help in providing support to the body.
- **(d) Mention one functional difference between A and B.**

    - Bone provides rigid structural support, while cartilage provides flexible support and cushioning.
- **(e) Write one specific location of each (A and B) in the human body.**
    - A (Cartilage): Tip of the nose. B (Bone): Bones of arms/legs.

4. (a) What are the two kinds of fibrous connective tissue?

- Tendons and Ligaments.
- **(b) Write one important function of each.**
    - **Tendons:** Connect muscle to bone. **Ligaments:** Connect bone to bone and hold them in position.
- **(c) Draw a neat diagram...**
    - *(Note: Practice sketching this from your textbook! Look for the bundle of fibres connected by areolar tissue.)* 🖍️

5. Sandesh observed a slide containing the longitudinal section of a plant tissue (Xylem)...

- **(a) Identify and name the kind of tissue shown. Is it simple or complex?**
    - Xylem (Vascular tissue). It is a **complex permanent tissue**.
- **(b) Distinguish between simple permanent tissue and complex permanent tissue.**
    - Simple tissues are made of only one type of cell. Complex tissues are made of more than one type of cell working together as a unit.
- **(c) Write one important role of the above mentioned tissue.**
    - Provides a passage for the upward movement of water and dissolved materials absorbed from the roots.

🧩 F. ASSERTION - REASON TYPE QUESTIONS

(Codes: (a) Both True, R explains A. (b) Both True, R doesn't explain A. (c) A True, R False. (d) A False, R True)

- **(i) A: Lateral meristem is responsible for increase in girth. R: Cambium is located near tips...**
    - **Answer: (c) A is true but R is false.** *(Reason: Cambium is lateral, apical is at tips).*
- **(ii) A: Collenchyma composed of long narrow cells with lignin. R: Collenchyma stores food.**
    - **Answer: (d) A is false but R is true.** *(Reason: Sclerenchyma has lignin, Collenchyma is thickened with pectin/cellulose).*
- **(iii) A: Tracheids are made up of elongated dead cells... R: Tracheids provide mechanical support...**
    - **Answer: (a) Both A and R are true and R is the correct explanation of A.**
- **(iv) A: Lymph is a fluid connective tissue which does not contain any blood cells. R: Lymph is concerned with transportation and immunity.**
    - **Answer: (d) A is false but R is true.** *(Reason: Lymph DOES contain white blood cells).*
- **(v) A: Cardiac muscles are striated, uninucleated and branched... R: Cardiac muscles can contract without outside stimulation and do not get tired easily.**
    - **Answer: (b) Both A and R are true but R is not the correct explanation of A.** *(They are structurally true, and functionally true, but the function doesn't explain the structural shape).*
- **(vi) A: Xylem and phloem are two types of vascular tissues... R: Xylem transports water... phloem transports prepared food...**
    - **Answer: (a) Both A and R are true and R is the correct explanation of A.**

🔬 G. CASE STUDY BASED QUESTION

1. Samay is working on a school project for his biology class... He notices actively dividing cells...

- **(a) Name the type of tissue observed by Samay that has the ability to divide repeatedly...**
    - Meristematic tissue.
- **(b) Write two specific locations in the plant body where the above mentioned tissue can be found.**
    - Tips of roots and tips of stems (or below the bark).
- **(c) State any two main characteristic features of the cells of this tissue.**
    -
        1. Vacuoles are almost absent. 2. Cells are tightly packed with no intercellular space.
- **(d) What are the 2 main categories of this tissue?**
    - Apical (or Terminal) meristem and Lateral (or Cambium) meristem.